higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N4

30 JULY 2019

This marking guideline consists of 8 pages.

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NOTE: √ = ½ mark
✓ = 1 mark

QUESTION 1

1.1 Workshop layout is the arrangement, storage and planning of different elements of a workshop. (3)

1.2
- Storage facilities
- Supply of services
- Transport routes and aisles
- Handling of workpieces
- Carrying capacity of the floor
- Space around and above each machine (Any 5 × 1) (5)

1.3 Excessive paint spray (1)
- The viscosity of the paint is too low.
- The atomising air pressure is too high.
- The distance between the spray gun and the workpiece is too great.

(Any TWO) (2)

Uneven spray-painting (1)
- There is insufficient air pressure.
- The spray nozzle is no longer effective because of obstructions, damage or a faulty adjustment.
- Blocked pipes reduce the paint supply. (Any TWO) (2)

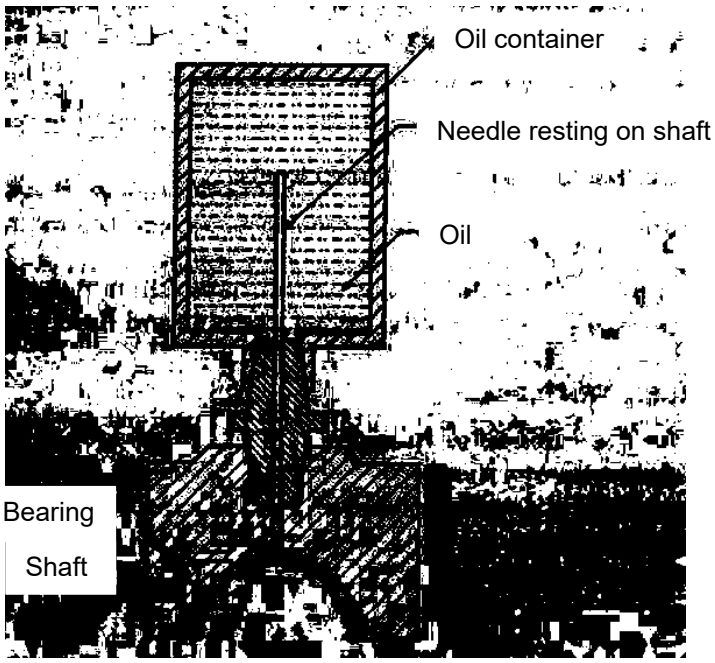
Sagging surface (1)
- The spray gun moves too slowly.
- Too much paint is applied to the surface.
- The spray gun is held too close to the surface.
- The paint is too thin. (Any TWO) (2)

Speckle or orange-peel effect (1)
- Wrong thinners or solvents are used.
- Paint is not mixed properly before use.
- Surface is prepared incorrectly.
- The air pressure is wrong. (Any TWO) (2)

1.4
- Minimal frictional resistance
- Strength
- Resistance to wear
- Thermal conductivity
- Resistance to corrosion (5)

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1.5



(5)
[30]

QUESTION 2

2.1
$$v=\frac{\pi(D+t)N}{60}$$
$$=\frac{\pi(0,6+0,02)\times 600}{60}\ \checkmark$$
$$=19,478\ m/s\ \checkmark$$
$$T_C=mv^2$$
$$=0,5(19,478)^2$$
$$=189,696\ N\ \checkmark$$
(3)

2.2
$$T_1=\sigma.w.t$$
$$=4\times 10^6\times 0,155\times 0,02\ \checkmark$$
$$=12\ 400\ N\ \checkmark$$
(2)

2.3
$$\frac{T_1-T_C}{T_2-T_C}=e^{\mu\theta}$$
$$\frac{12400-189,696}{T_2-189,696}=e^{0,2\times 2,792}\ \checkmark$$
$$\frac{12210,304}{T_2-189,696}=1,747\ \checkmark$$
$$12210,304=1,747T_2-331,398\ \checkmark$$
$$T_2=\frac{12541,702}{1,747}\ \checkmark$$
(5)

$\therefore T_2 = 7178,994\,N$ ✓

2.4 $P=(T_1-T_2)v$
$=(12\ 400-7178,994)19,478$ ✓
$=101694,755\ W$ ✓ (2)

2.5 $P_i=\frac{P_o}{\eta}$
$P_i=\frac{101694,755}{0,8}$ ✓
$=127118,444\ W$ ✓ (2)
[14]

QUESTION 3

3.1 3.1.1 $Cutting\ Area=Feed/\mathrm{Re}v\times Cutting\ depth$
$=0,5\times3,5$
$=1,75\ mm^2$ ✓
$F=PxA$
$=1000x1,75$
$=1\ 750N$ ✓
$v=\frac{\pi.D.N}{60}$
$=\frac{\pi\times0,5x30}{60}$
$=0,785\ m/s$ ✓
$P_o=F\times v$
$=1750\times0,785$
$=1374,446\ W$ ✓ (4)

3.1.2 $P_i=\frac{P_o}{\eta}$
$P_i=\frac{1374,446}{0,85}$ ✓✓
$=1616,996\ W$ ✓ (3)

3.2 3.2.1 $\mu=\frac{F_f}{N_R}$
$F_f=\mu\times N_R$
$=0,07\times200\times9,81$ ✓✓
$=137,34N$ ✓ (3)

3.2.2

$$\mu = \frac{F_f}{N_R}$$

$$\mu = \frac{137,34 + (137,34\,x\,0,3)}{200 \times 9,81} \checkmark\checkmark$$

$$\mu = \frac{178,542}{1\,962} \checkmark$$

$$= 0,091 \checkmark$$

(4)
[14]

QUESTION 4

4.1 4.1.1

$$C = \frac{m}{2}(T_A - T_B)$$

$$45 = \frac{1,5}{2}(T_A + 30) \checkmark$$

$$90 = 1,5\,T_A - 45) \checkmark$$

$$T_A = \frac{135}{1,5}$$

$$= 90\,teeth \checkmark$$

4.1.2 $PCD_A = mxT_A$

$$= 1,5 \times 90 \checkmark\checkmark$$

$$= 135\ mm \checkmark$$

(2 × 3) (6)

4.2

CONDITION	PINION A	GEAR WHEEL B	ARM C
Fix arm C and rotate gear B by +1 rev	$\frac{80}{-20} = -4$ √	$= +1$ √	*0* √
Multiply by x and add y	$-4x + y$ √	$x + y$ √	$+ y$ √
$N_A = ?$ $N_B = 0\,r/\text{min}$ $N_C =$	$N_A =$?√	$N_B = 0$ √	$N_C = +100$ √

$$x + y = 0 \ldots\ldots 1$$

$$y = 100 \ldots\ldots 2$$

Substitute …2 in ….1

$$x = -y \surd$$

$$x = -100 \checkmark$$

$$N_A = -4x + y$$

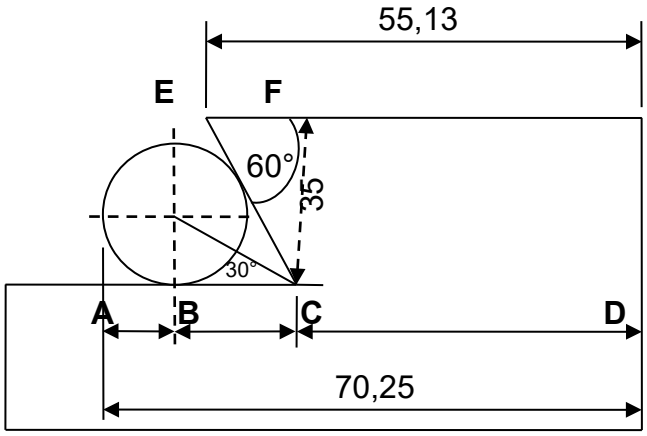
$$N_A = -4(-100) + 100 \checkmark$$

(8)

$N_A = +500\ r/\min\ clockwise$ ✓ **[14]**

QUESTION 5

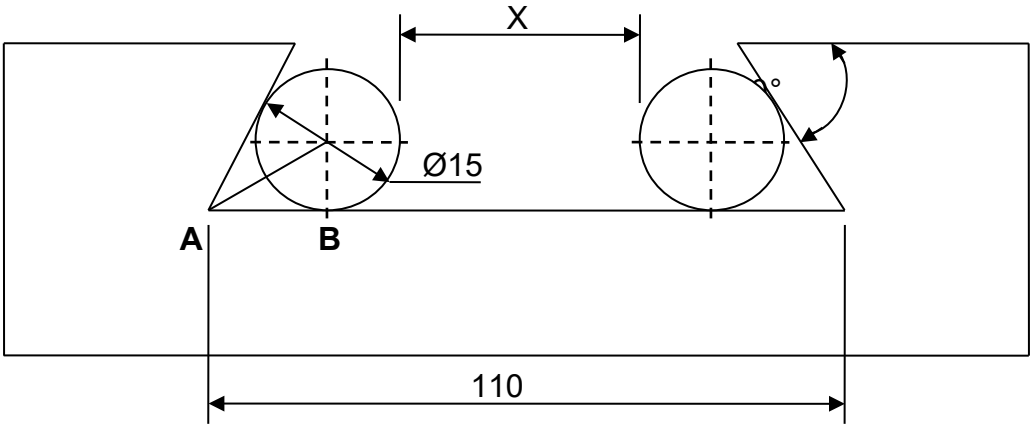
5.1



$$AD = AB + BC + CD$$
$$AB = r$$
$$\tan 30° = \frac{r}{BC}$$
$$0,577 = \frac{r}{BC}$$
$$BC = 1,732r \checkmark$$
$$CD = 55,13 - EF$$
$$\tan 60° = \frac{35}{EF}$$
$$EF = \frac{35}{\tan 60°} \checkmark$$
$$EF = 20,207 \checkmark$$
$$CD = 55,13 - 20,207$$
$$CD = 34,923 \checkmark$$
$$AD = AB + BC + CD$$
$$70,25 = r + 1,732 + 34,923$$
$$35,327 = 2,732r \checkmark$$
$$r = \frac{35,327}{2,732}$$
$$r = 12,931 \checkmark$$
$$D = 25,862\ mm \checkmark$$

(7)

5.2



$$X = 110 - 2\,AB - 2\,r \checkmark$$
$$r = 7{,}5 \checkmark$$
$$\tan 20° = \frac{7{,}5}{AB}$$
$$AB = \frac{7{,}5}{\tan 20°} \checkmark$$
$$AB = 20{,}606 \checkmark$$
$$X = 110 - 2\,AB - 2\,r$$
$$X = 110 - 2 \times 20{,}606 - 2 \times 7{,}5 \checkmark$$
$$X = 110 - 56{,}212 \checkmark$$
$$X = 53{,}788\ mm \checkmark$$

(7)
[14]

QUESTION 6

6.1
$$v_t = \sqrt{2\,gh}$$
$$= \sqrt{2 \times 9{,}81 \times 35{,}5}$$
$$= 26{,}391\ m/s \checkmark$$

$$Q_t = v_t \times A_t$$
$$= 26{,}391 \times \frac{\pi (0{,}032)^2}{4} \checkmark$$
$$= 0{,}0212\ m^3/s \checkmark$$
$$C_d = \frac{Q_a}{Q_t}$$
$$= \frac{0{,}0128}{0{,}0212}$$
$$= 0{,}605 \checkmark$$

(4)

6.2

$$C_c = \frac{A_a}{A_t}$$
$$= \frac{(0,027)^2}{(0,032)^2} \checkmark\checkmark$$
$$= 0,712 \checkmark$$

(3)

6.3

$$v_a = \frac{Q_a}{A_t}$$
$$= \frac{0,0128}{5,725 \times 10^{-4}} \checkmark$$
$$= 22,358\ m/s \checkmark$$
$$C_v = \frac{v_a}{v_t}$$
$$= \frac{22,358}{26,391} \checkmark$$
$$= 0,847 \checkmark$$

(4)

6.4

$$h_2 = \frac{(v_2)^2}{2g}$$
$$= \frac{22,358^2}{2 \times 9,81} \checkmark$$
$$= 25,478\ m \checkmark$$
$$Head\ loss = h_1 - h_2$$
$$= 35,5 - 25,478$$
$$= 10,022\ m \checkmark$$

(3)
[14]

TOTAL: 100

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